

M. SYARAFUDDINIL UQAILI

ELECTRICAL ENGINEERING

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EDUCATION

Electrical Engineering, Syiah Kuala University **2020-2024**

Graduated as an Engineering Bachelor with GPA 3.42 and good predicate. Taken a vocational field in control engineering. High scored in programming and robotics subject. Integrated signal processing with CNN as a deep learning method in thesis to classify normal and abnormal heart conditions.

RESEARCH INTEREST

Artificial Intelligence, Embedded System, Autonomous System, Internet of Things, Robotics

RESEARCH EXPERIENCE

Undergraduate Thesis **Mar - Aug 2024**
Detection of Diabetes Using Photoplethysmography Signals with a Deep Learning Approach, Syiah Kuala University

Programmed and designed wiring layouts of microcontrollers using Arduino IDE. Configured temperature, humidity, and pH sensors for monitoring the growth conditions of pakcoy vegetables, which resulting in 90% accuracy from the calibration.

PROFESSIONAL EXPERIENCE

Technical Division Staff **Jul - Aug 2022**
PT. PIM (Pupuk Iskandar Muda)

- Assisted in PLC troubleshooting and system optimization
- Supported proposal drafting for industrial automation prototype
- Designed and drafted comprehensive wiring diagrams using AutoCad.

PLC Programming – Schneider Machine Expert Basic **Nov 2023**
Creative Station

- Completed intensive training on PLC programming, ladder logic implementation, and industrial automation system configuration using Schneider Machine Expert Basic.

Competency-Based Training (CBT) – Residential Air Conditioning Maintenance Program **Apr - Mei 2025**

Vocational and Productivity Training Center Banda Aceh

- Developed understanding of refrigeration cycles and AC system operation.
- Conducted routine maintenance and troubleshooting of residential air conditioning units.
- Utilized electrical measuring instruments and service tools for system inspection and diagnostics.

AWARD & HONOR

Smart Agriculture System: Kebonku **Dec 2022**
Top 150 Social Project Innovillage Competition

- Developed Python and Arduino-based programs for real-time soil quality monitoring
- Implemented sensor integration to ensure reliable and continuous environmental data collection
- Configured and managed Raspberry Pi as a central hub for data acquisition and processing

Smart Healthcare System: Integrated Deep Learning with IoT
Funding Recipient on Student Creativity Program Competition

Nov 2023

- Developed front-end and back-end architecture for a real-time medical diagnostic website
- Configured hardware wiring and data integration to enable live heart rate monitoring
- Implemented real-time data transmission between sensor system and web platform

SOCIAL INVOLVEMENT

Autonomous Robotic

Feb 2022 - Feb 2023

Electrical Engineering Robotic Community, Syiah Kuala University

Participated in developing an autonomous robot program that can perform several simple movements according to commands while minimizing errors. This robot successfully became a finalist in a national robotics competition.

Education and Human Resource Development Division Member

Feb 2022 - Feb 2023

Himpunan Mahasiswa Teknik Elektro (HIMATEKTRO), Syiah Kuala University

Conducted research on academic trends and student preferences, increased student engagement in academic and extracurricular activities, and organized seminars and workshops to support institutional programs.